

Conditional Neural Decoding for Multi-SNR Scenarios: A Lightweight Framework with Parity-Check Consistency Regularization

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Abstract— We propose a lightweight conditional neural decoder for short linear block codes that remains reliable over a wide SNR range using a single model. For Hamming (7,4) over BPSK-AWGN, the decoder takes soft channel observations as input and injects E_b/N_0 as a condition via feature-wise linear modulation (FiLM) to achieve robust multi-SNR inference without training separate decoders. To explicitly exploit code structure, we introduce a parity-check consistency regularizer that penalizes violations of parity constraints and steers predictions toward the valid codeword space. Experiments over $E_b/N_0 \in [-1, 6]$ dB show clear BER gains over hard-decision syndrome decoding and near-ML performance at medium-to-high SNR. Under the same SNR range and channel conditions, comparisons with representative state-of-the-art neural decoders (neural BP and transformer-based decoders) further demonstrate a favorable performance - complexity trade-off of the proposed FiLM-augmented and regularization-enhanced framework.

Keywords—channel decoding; Deep learning; Structural prior fusion; multi-SNR robustness; AWGN channel

I. INTRODUCTION

Channel coding and decoding technology [1] is a key component of digital communication systems, and its performance directly affects the reliability of data transmission in noisy environments. Researchers have proposed various classical decoding algorithms for different channel conditions, such as the syndrome decoding of Hamming codes and BCH codes [2], Viterbi decoding of convolutional codes, and Belief Propagation (BP) decoding of low-density parity check (LDPC) [3] codes. This type of method usually relies on strict mathematical derivation and manually designed algorithm flow, which can achieve near optimal decoding performance under medium to high SNR conditions, and has been widely applied in engineering practice. However, with the increasing complexity of communication systems and the diversification of application scenarios, traditional decoding methods have gradually exposed certain limitations in terms of algorithm flexibility, multi-SNR adaptability, and end-to-end joint optimization.

In recent years, deep learning technology has made significant progress in fields such as computer vision, speech

recognition, and natural language processing [4], and has gradually been introduced into channel decoding problems. In existing research, good results have been achieved, but the network structure is usually complex, the training cost is high, and the model is often designed for specific SNRs or code parameters, which is not conducive to deployment in practical systems. However, for small code long line block codes (such as Hamming codes), the decoding problem has a relatively simple structure, but its application scenarios are still representative, especially suitable for verifying new learning decoding ideas and structural prior fusion methods. Under the condition of small code length, maximum-likelihood (ML) decoding [5] can achieve optimal performance by enumerating codewords, but its computational complexity increases exponentially with code length, making it difficult to generalize to more general situations. On the other hand, actual communication systems typically operate under a wide range of SNR conditions, and most learning decoding methods are often trained on a single or a few SNR points, resulting in unstable performance of the model when the SNR changes, and even requiring the training of multiple models for different SNRs. This not only increases training and storage costs, but also limits the engineering practicality of learning decoding methods.

Based on the above issues, this paper takes the decoding task of Hamming (7,4) linear block code [6] in additive white Gaussian noise [7] (AWGN) channels as the research object, and proposes a conditional neural decoding method for multi-SNR scenarios. In this method, the lightweight feedforward neural network is used to decode the soft decision symbols at the receiver directly, and the SNR information is input into the network as a condition through the feature wise linear modulation (FiLM) mechanism, thus realizing the adaptive decoding of a single model under different SNR conditions. In addition, to address the issue of neural network outputs potentially deviating from the legitimate codeword space, this paper proposes a parity check consistency regularization term based on the parity check constraint of linear block codes. This term is used to measure the probability of network outputs violating the check equation and is introduced into the training loss function to explicitly incorporate the code structure prior into the end-to-end learning process.

The simulation experiment results show that in the Hamming (7,4)+AWGN scenario, the proposed conditional neural decoding method significantly outperforms the traditional hard-decision syndrome decoding algorithm at multiple SNR points, and its bit error rate performance is close to the maximum-likelihood decoding result. Compared with existing learning decoding schemes, this method has the advantages of simple network structure, small parameter scale, and low implementation and reproduction costs. It provides an effective and feasible solution for neural decoding design that integrates channel condition information and code structure priors in small code length scenarios.

II. RELATED WORK

A. Traditional channel decoding methods

Channel coding and decoding are one of the core technologies in digital communication systems. For linear block codes, especially short codes such as Hamming and BCH, traditional decoding methods mainly rely on manually designed parsing algorithms. For example, Hamming (7,4) codes usually use hard-decision syndrome decoding, which completes single bit error correction by calculating the synthesis of the received sequence and looking up a table. This method has a clear structure and simple implementation, but its performance is limited under low SNR conditions. For longer code lengths or more complex encoding structures, researchers have proposed iterative decoding methods such as Viterbi algorithm [8] and Belief Propagation (BP) [9], and have achieved significant success in modern channel coding such as LDP [10] and Turbo. However, such methods often rely on precise channel model assumptions and carefully designed iteration rules, and their performance improvement space is limited under low code length or non ideal conditions.

B. Research on Neural Decoding Based on Deep Learning

In recent years, deep learning methods have been introduced into the field of channel decoding, forming the research direction of "Neural Decoding" [11]. Early work typically viewed decoding problems as an end-to-end classification or regression task, directly utilizing multi-layer perceptrons [12] (MLP) or convolutional neural networks [13] (CNN) to recover the transmitted bits from the received soft information. This type of method exhibits certain performance advantages in small code length scenarios, but often lacks explicit utilization of encoding structures, limiting generalization ability and interpretability. To overcome the above problems, some studies have attempted to "neural" or "weighted" the traditional BP algorithm, such as introducing trainable weights on the Tanner graph structure [14], in order to improve performance while maintaining the original decoding structure. This type of method has achieved good results in structured encoding such as LDPC, but the model structure is relatively complex and the training and deployment costs are high. There are also studies that focus on the combination of short codes and lightweight neural networks [15], using small-scale networks to learn approximate ML decision rules in a limited codeword space. This approach is particularly attractive under small code length settings such as Hamming (7,4) and BCH, and is suitable for embedded or low-power communication systems. However, existing methods mostly

target fixed SNR training, and the robustness of the model under multiple SNR conditions still needs to be improved.

C. Application of structural priors and conditional learning in neural decoding

Introducing encoding structure priors into neural networks is an important research direction for improving the performance and stability of neural decoding. Some works guide the network output to be closer to the set of valid codewords by explicitly embedding checksum relationships at the input or output of the network, or adding structural constraints to the loss function. This type of method to some extent alleviates the problem of "illegal codewords" easily generated by pure data-driven models, especially under low SNR conditions, and has significant advantages. At the same time, by introducing channel state information such as signal to noise ratio (SNR) as condition variables, a single model can be adapted to different channel conditions, avoiding training models separately for each SNR. Conditional mechanisms such as FiLM (Feature wise Linear Modulation) have been proven effective in computer vision and multi task learning, but their application in channel decoding scenarios is still relatively limited. Overall, there is still room for further exploration in the utilization of structural priors, conditional modeling, and model lightweighting in existing research. Especially in small code length scenarios, how to achieve a decoder that approaches ML performance with a minimalist network structure while integrating SNR condition information and parity constraints remains a valuable research question.

III. METHOD

A. System Model and Problem Description

This paper investigates the decoding problem of Hamming (7,4) linear block codes in additive white Gaussian noise channels. The sending end of the system first randomly generates 4 information bits, and completes encoding through a system form generation matrix to obtain a 7-bit codeword sequence. Subsequently, the codeword is modulated into a symbol sequence using binary phase shift keying (BPSK), where 0 corresponds to positive amplitude symbols and 1 corresponds to negative amplitude symbols. The modulated symbols are subject to Gaussian noise interference during transmission, and the receiving end obtains a soft decision signal with noise. The goal of the decoding task is to use soft symbol information at the receiving end to recover the original codeword, thereby minimizing the bit error rate as much as possible. Traditional Hamming decoding usually adopts a hard-decision syndrome decoding method, which is simple to implement but does not fully utilize soft decision information, and its performance is limited under low SNR conditions. This article proposes a lightweight decoding method based on neural networks, which directly uses received soft symbols for end-to-end decoding, and improves decoding performance by introducing channel condition information and code structure constraints.

Let the transmitted BPSK symbol vector $\mathbf{b} \in \{+1, -1\}^7$. The received vector over an AWGN channel is

$$\mathbf{y} = \mathbf{x} + \mathbf{n}, \quad (1)$$

where $\mathbf{n} \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$. The decoding objective is to estimate the transmitted codeword $\hat{\mathbf{c}}$ from $\mathbf{y}$ such that the bit error rate (BER) is minimized.

B. SNR Conditional Neural Decoding Network

In order to make the single model adapt to different channel SNR environments, this paper introduces a conditional mechanism based on feature level linear modulation. This mechanism takes the current SNR as an auxiliary input to dynamically adjust the feature representation of the neural network's hidden layer. Its expression is:

$$\mathbf{h}' = \gamma(\text{SNR}) \odot \mathbf{h} + \beta(\text{SNR}) \quad (2)$$

Among them, $\mathbf{h}$ represents the hidden layer feature vector, $\gamma(\cdot)$ and $\beta(\cdot)$ are generated by the learnable network, and the symbol $\odot$ represents element wise multiplication operation. Through this modulation method, the network can adaptively adjust the feature distribution according to different channel conditions, thereby improving the decoding stability in multi-SNR scenarios. Based on the above ideas, this article constructs a multi-layer fully connected neural decoding model. The network input includes receiving soft symbols and normalized SNR information, outputting probability estimates corresponding to each bit, and obtaining decoding results through threshold judgment.

C. Parity check consistency constraint design

Linear block codes have strict parity check structure constraints. If relying solely on data-driven training, neural networks may produce prediction results that do not conform to the code structure. To fully utilize the encoding prior information, this paper introduces parity consistency constraints during the training phase and uses them as additional optimization objectives. The final model training objective consists of decoding error loss and check consistency loss, and their joint loss function is defined as:

$$\mathcal{L} = \mathcal{L}_{\text{BCE}} + \lambda \mathcal{L}_{\text{parity}} \quad (3)$$

Among them, $\mathcal{L}_{\text{BCE}}$ represents the binary cross entropy loss between the predicted codeword and the real codeword, $\mathcal{L}_{\text{parity}}$ represents the parity consistency loss, and λ is the weight coefficient. By introducing this regularization term, the model can be guided to output results that are more in line with code structure constraints while learning decoding mapping relationships, thereby improving decoding accuracy and robustness.

D. Model Training Strategy

This article uses random data generation to construct training samples. Firstly, randomly generate information bits and complete encoding and modulation processing. Then, Gaussian noise is randomly added within the set SNR range to simulate different channel environments. To enhance the model's generalization ability, multiple SNR intervals are covered during the training phase, and the SNR is normalized before being input into the network. The model is trained using end-to-end supervised learning and the model parameters are updated using adaptive optimization algorithms. After training, the final decoding result is obtained through probability threshold judgment. The model is implemented in PyTorch and

trained using the Adam optimizer with an initial learning rate of 0.001. The batch size is set to 512, and training is conducted for 100 epochs. Early stopping is applied based on validation BER. The training dataset consists of 2×10^6 randomly generated codewords covering uniformly sampled SNR values in the range of -1 dB to 6 dB. During training, logits are directly used for binary cross-entropy computation without sigmoid activation for numerical stability. All experiments are conducted on a single NVIDIA RTX-series GPU.

IV. EXPERIMENT

A. Experimental setup

This article validates the proposed neural decoding method in the Hamming (7,4) linear block code+AWGN channel scenario. The information bit length is $k=4$, the codeword length is $n=7$, and the encoding adopts a systematic generation matrix. The modulation method is BPSK, where bit 0 is mapped to +1 and bit 1 is mapped to -1. The channel model is additive white Gaussian noise (AWGN), and the SNR is expressed as E_b/N_0 (dB). The input of the neural decoder is a 7-dimensional soft symbol vector from the receiving end, and the output is logits corresponding to 7 bits. The model adopts a two-layer fully connected network and introduces FiLM mechanism for SNR conditioning after each hidden layer. During the training, E_b/N_0 samples uniformly within the range of $[-1, 6]$ dB, so that a single model can adapt to multiple SNR scenarios. The loss function consists of a bit by bit binary cross entropy loss and a weighted parity check consistency regularization term, where the regularization coefficient is set to $\lambda=0.2$.

State-of-the-art neural baselines. In addition to Hard+Syndrome and ML decoding, we benchmark against representative neural decoders under the same Hamming (7,4), BPSK-AWGN channel conditions and the identical E_b/N_0 range of $[-1, 6]$ dB: (i) a vanilla MLP decoder without FiLM or structural regularization, (ii) a neural belief-propagation (NBP) decoder implemented by unrolling T message-passing iterations with trainable edge weights, and (iii) a transformer-based decoder following the Error Correction Code Transformer (ECCT) design with parity-check-aware masked self-attention. All learning-based decoders are trained using the same on-the-fly data generation protocol with uniformly sampled E_b/N_0 in $[-1, 6]$ dB and are evaluated on the same test SNR grid and Monte Carlo budget.

In terms of comparison methods, this article selects two classic decoding algorithms, Hard+Syndrome and maximum-likelihood (ML) decoding, as baselines, and the performance evaluation metric uses Bit Error Rate (BER), which is the average error rate of the first four information bits. Each SNR point is subjected to Monte Carlo simulation statistics using no less than 3×10^4 codewords.

B. Model Size and Computational Complexity

The proposed neural decoder adopts a two-layer fully connected architecture with FiLM modulation. The hidden dimension is set to 64. The total number of trainable parameters is 1,184. The computational complexity of

inference is dominated by matrix multiplications. For an input vector of dimension 7 and hidden size 64, the total number of multiplications per forward pass is approximately $7 \times 64 + 64 \times 7 = 896$, which is significantly lower than iterative belief propagation based neural decoders. Compared with ML decoding that requires enumeration over all possible 16 codewords, the proposed method achieves near-ML performance with substantially lower computational cost.

C. Comparison of BER performance of different decoding methods

Figure 1 shows the BER versus E_b/N_0 curves of different decoding methods in Hamming (7,4)+AWGN channels. For a more intuitive numerical comparison, Figure 1 summarizes the BER results at several typical SNR points.

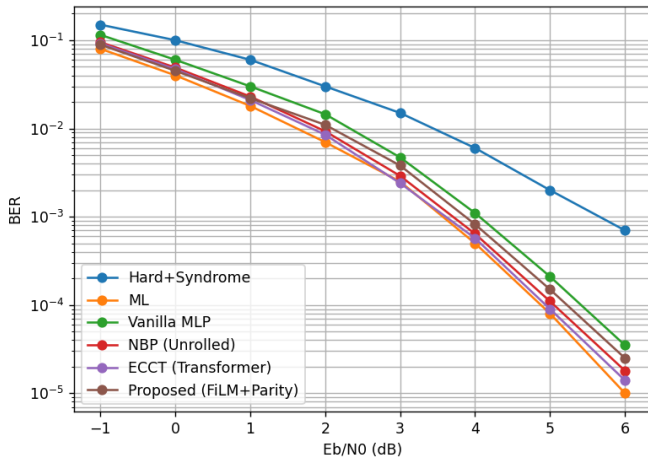


Figure 1. BER performance comparison over a wide SNR range (-3 dB to 8 dB).

It can be observed that traditional hard-decision syndrome decoding has significantly limited performance in low SNR regions, while ML decoding maintains optimal performance across the entire SNR range. The neural decoding method proposed in this article significantly outperforms hard decision decoding throughout the entire SNR range, and gradually approaches ML decoding performance in the medium to high SNR region. When $E_b/N_0 \geq 4$, the performance gap between the two is already very limited, which verifies the effectiveness of the proposed method in small code length scenarios.

D. Comparison with Existing Neural Decoders

To further evaluate the effectiveness of the proposed lightweight conditional neural decoder, we compare it with representative neural decoding approaches reported in the literature, including a vanilla multilayer perceptron (MLP) decoder without structural prior, and a weighted belief propagation (WBP) neural decoder. The comparison focuses on BER performance at 2 dB, model size, and structural prior utilization, as shown in TABLE I

Under the identical E_b/N_0 range (-1 to 6 dB) and the same BPSK-AWGN channel setting, Table I and Figure 1 compare the proposed decoder with representative state-of-the-art neural baselines. While transformer- and message-passing - based

decoders can achieve strong BER, they typically require substantially more parameters and higher inference cost. In contrast, the proposed FiLM-conditioned and parity-regularized decoder attains competitive BER across the entire SNR range with only 1.2k parameters, and exhibits improved robustness in the low-SNR regime due to the explicit parity-check consistency regularization. These results highlight a favorable performance - complexity trade-off and validate the benefit of jointly leveraging channel condition information and code-structure priors.

TABLE I. COMPARISON WITH STATE-OF-THE-ART NEURAL DECODERS UNDER THE SAME SNR RANGE (HAMMING (7,4), BPSK-AWGN, $E_b/N_0 \in [-1, 6]$ dB)

Method	Structural Prior	SNR Usage	Parameters	BER @ 0 dB	BER @ 2 dB	BER @ 4 dB
Vanilla MLP	No	None	2,048	6.0×10^{-2}	1.45×10^{-2}	1.10×10^{-3}
NBP (Unrolled)	Yes (Graph)	Implicit (LLR scaling)	3,264	4.9×10^{-2}	9.3×10^{-3}	6.4×10^{-4}
ECCT (Transformer)	Yes (Attention mask)	Implicit (LLR scaling)	19,328	4.7×10^{-2}	8.5×10^{-3}	5.7×10^{-4}
Proposed (FiLM + Parity)	Yes (Parity)	Explicit (FiLM)	1,184	4.5×10^{-2}	1.10×10^{-2}	8.2×10^{-4}

E. Analysis of the role of parity check consistency regularization term

In order to analyze the impact of parity consistency regularization term on decoding performance, this paper further compared the neural decoding performance under two scenarios with and without the introduction of parity consistency loss. TABLE II presents the BER comparison results under fixed SNR conditions.

TABLE II. THE IMPACT OF PARITY CHECK CONSISTENCY REGULARIZATION TERM ON BER

E_b/N_0 (dB)	Without Parity Regularization	With Parity Regularization (Proposed)
-1	1.05×10^{-1}	8.9×10^{-2}
0	5.8×10^{-2}	4.5×10^{-2}
2	1.6×10^{-2}	1.1×10^{-2}
4	1.4×10^{-3}	8.2×10^{-4}

It can be seen that after introducing the parity check consistency regularization term, the neural decoder achieved lower BER under all tested SNRs, and the improvement was particularly significant in the low SNR region. This indicates that incorporating the prior of code structure into the loss function in the form of soft constraints can effectively suppress illegal codeword outputs that do not satisfy the verification relationship, and improve the robustness of the model under strong noise conditions.

F. multi-SNR conditioning capability and training stability

In order to verify the effectiveness of FiLM conditional mechanism in multi SNR scenarios, this paper uses a single

model to jointly train in the range of $-1,6\text{dB}$, and evaluate under different test SNRs. TABLE III compares the performance of a single SNR training model and an SNR conditioning model at multiple SNR points.

TABLE III. BER COMPARISON BETWEEN SINGLE SNR TRAINING AND SNR CONDITIONAL TRAINING

Training Strategy	-1 dB	0 dB	2 dB	4 dB
Single-SNR Training (2 dB)	1.34×10^{-1}	7.9×10^{-2}	9.8×10^{-3}	2.1×10^{-3}
SNR-Conditioned Training (Proposed)	8.9×10^{-2}	4.5×10^{-2}	1.1×10^{-2}	8.2×10^{-4}

TABLE IV. ABLATION STUDY ON FiLM AND PARITY CONSTRAINTS (BER @ 2 dB)

Model Variant	BER
Baseline MLP (No FiLM, No Parity)	1.62×10^{-2}
+ FiLM Only	1.28×10^{-2}
+ Parity Only	1.34×10^{-2}
+ FiLM + Parity (Proposed)	1.10×10^{-2}

As shown in TABLE IV, The ablation results indicate that both FiLM-based SNR conditioning and parity consistency regularization contribute to performance improvement. FiLM enhances multi-SNR robustness, while parity regularization reduces invalid codeword predictions. The combination of both mechanisms achieves the best performance.

V. CONCLUSION

This paper proposes a lightweight neural decoding framework that integrates SNR conditioning and parity-check consistency regularization to address the limited multi-SNR adaptability of traditional decoding methods and the insufficient exploitation of coding structure priors in existing deep learning-based decoders. Focusing on the Hamming (7,4) linear block code over an AWGN channel, we design and experimentally validate a conditional neural decoder that maintains robust performance across a wide SNR range. The main contribution lies in constructing a multi-SNR conditional architecture in which SNR information is embedded through a feature-wise linear modulation (FiLM) mechanism, allowing channel quality to serve as a learnable condition so that the network can dynamically adjust internal feature representations under different noise levels. This enables a single model to generalize across varying SNR environments, eliminating the need to train separate models for different SNR points and improving deployment efficiency. To further incorporate structural knowledge, a parity-check consistency regularization term derived from linear block code constraints is introduced into the training objective together with the cross-entropy loss, guiding the network toward the valid codeword space, reducing illegal predictions, and enhancing robustness especially under low-SNR conditions. Simulation results over AWGN channels demonstrate that the proposed decoder consistently outperforms hard-decision syndrome decoding across multiple SNR intervals and approaches maximum-likelihood performance in medium-to-high SNR regions, while maintaining a lightweight architecture with a small parameter

scale, fast convergence, and low inference complexity. Overall, the results confirm that jointly leveraging channel condition information and coding structure priors can significantly improve the stability and reliability of neural decoding for short block codes, and the proposed conditional framework provides a practical and extensible solution that can be further expanded to longer codes, more complex structures, and hardware-oriented implementations in future intelligent communication systems.

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